## 1. Overview for AudioPeakRMSChecker.ps1

This script analyzes the audio tracks inside a movie file (MKV, MP4). It checks for:
**Peak** (dB) loudest moment in the audio.
**RMS** (dB) average loudness over time.
**Crest Factor** (dB) difference between Peak and RMS
**Clipping** (number of samples that hit or exceed 0 dB)

The function is to give you a quick PASS/FAIL result so you can tell whether the audio is clean, safe, and ready for encoding, downmixing, or playback. It supports common multichannel formats including TrueHD, AAC, EAC3/DDP.

This tool does **not** modify audio. It only analyzes it and reports the results.

## 2. Requirements

To run AudioPeakRMSChecker, you need:

- PowerShell 7.6
- FFmpeg 8.1 (ffmpeg and ffprobe must be placed in the same folder as the script.)

Official FFmpeg download page: https://ffmpeg.org/download.html

## Quick Start (Windows 11)

1. Place `AudioPeakRMSChecker.ps1`, `ffmpeg.exe`, and `ffprobe.exe` in the same folder.
2. Run the script:

```
pwsh -ExecutionPolicy Bypass -File .\AudioPeakRMSChecker.ps1 ".\YourMovie.mkv"
```

ℹ️ The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

## macOS / Linux

1. Place the script, `ffmpeg`, and `ffprobe` in the same folder.
2. Make the binaries executable (first time only):

```
chmod +x ffmpeg ffprobe
```

1. Run:

```
pwsh -File ./AudioPeakRMSChecker.ps1 "./YourMovie.mkv"
```

## 3. Supported Audio Formats

AudioPeakRMSChecker analyzes the following audio formats:

| Codec Family | Formats Included |
|---|---|
| **TrueHD** | TrueHD 7.1 |
| **AAC** | AAC 7.1 |
| **EAC3 / DDP** | Dolby Digital Plus 5.1 |
| **DTS** | DTS-HD 7.1 |

The tool automatically detects:

- the codec
- the number of channels
- whether the track is a source (7.1) or downmix (5.1)
- whether the track is supported for analysis

Unsupported tracks are ignored safely.

## 4. How the Analysis Works

The script performs the following steps:

1. **Scans the movie file**

   It looks for audio tracks and identifies their codec and channel layout.
2. **Runs FFmpeg's "astats" filter**

   This filter measures loudness, peaks, clipping, and dynamic range.
3. **Collects the results**

   The script extracts the important numbers from FFmpeg's output.
4. **Checks for problems**

   It looks for:

   - peaks too close to 0 dB
   - audio that is too loud overall
   - low dynamic range
   - clipped samples
5. **Returns PASS or FAIL**

   PASS means the audio is clean and safe.
   FAIL means the audio may distort or has loudness issues.
6. **Saves a log file** (*.astats.txt)

   A text file is created with the full FFmpeg analysis for reference.

The entire process is automatic and requires no technical knowledge.

## 5. What Each Metric Means

AudioPeakRMSChecker reports four main metrics:

### Peak (dB)

- The loudest moment in the audio.
- Values **near 0 dB** mean the audio is very loud.
- Values **above 0 dB** indicate clipping.

### RMS (dB)

- The average loudness over time.
- Higher RMS (closer to 0 dB) means the audio is louder.
- Lower RMS (–20 dB to –35 dB) means more dynamic range.

### Crest Factor (dB)

- The difference between Peak and RMS.
- Higher crest = more dynamic, cinematic sound.
- Low crest = compressed or flat audio.

### Clipped Samples

- The number of samples that hit or exceed 0 dB.
- A few clips are usually harmless.
- Many clips indicate distortion.

These metrics help determine whether the audio is clean or risky.

## 6. PASS / FAIL Rules (Simple)

The script marks a track as **FAIL** if any of the following are true:

- **Peak > 0 dB**

  The audio is clipping.
- **RMS louder than –12 dB**

  The audio is extremely loud and may distort.
- **Crest < 6 dB**

  The audio is heavily compressed.
- **Clipped samples > 5**

Too many clipped peaks.

- **Peak ≥ 0 dB with any clipped samples**

  Even a single clipped sample triggers FAIL when the peak is at or above 0 dB.

If none of these conditions are met, the track is marked **PASS**.
PASS means:

- clean audio
- safe headroom
- healthy dynamics
- no distortion

FAIL means:

- the audio may distort
- the mix may be too loud
- the track may need correction

## 7. JSON Output Mode

AudioPeakRMSChecker can output results in **JSON format**, which is useful for:

- automation
- pipelines
- dashboards
- batch processing

Located at top of script. Line32
To **enable** JSON mode: $JsonMode = $true
To **disable** JSON mode: $JsonMode = $false

In JSON mode:

- No summary is printed
- No color output
- No interpretation text
- Only the JSON object is returned

## Exit Codes:

| number | description |
|---|---|
| **0** | Success |
| **1** | Input validation: file missing / no audio streams / no supported codecs / no valid 7.1 or DDP 5.1 combination |
| **2** | FFmpeg produced no stderr output (analysis aborted) |
| **3** | FFmpeg exited with non-zero exit code |
| **5** | FFmpeg exceeded maximum allowed runtime (timeout guardrail) |
| **6** | astats output missing required metrics (Peak / RMS / Peak count) |
| **7** | Peak level dB failed numeric parse |
| **8** | RMS level dB failed numeric parse |
| **9** | Peak count failed numeric parse |
| **10** | astats log file was not created on disk |
| **11** | astats log file is empty or smaller than 10 bytes |
| **12** | ffmpeg or ffprobe binary missing or not executable in the script folder |